FINAL EXAMINATION
SEMESTER II, SESSION 2019/2020

COURSE CODE	:	FSPM 0024
COURSE	:	CALCULUS
PROGRAMME	:	FOUNDATION UTM
DURATION	:	3 HOURS
DATE	:	10 NOVEMBER 2019

INSTRUCTION TO CANDIDATES:

ANSWER ALL QUESTIONS.

WARNING!

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This examination question consists of (6) printed pages only including this page

QUESTION 1 (15 MARKS)

a) A function $f(x)$ is defined by

$$f(x) = \begin{cases} a^2x - 2, & x \leq 1, \\ bx^2, & 1 < x \leq 2, \\ x^3, & x > 2, \end{cases}$$

where a and b are constants. Determine the values of a and b if $a > 0$, so that $f(x)$ is continuous at $x = 1$ and $x = 2$.

(6 marks)

b) Find the limit of each of the following:

i. $\displaystyle\lim_{x \to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$.

(4 marks)

ii. $\displaystyle\lim_{x \to \infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$.

(5 marks)

QUESTION 2 (15 MARKS)

a) If $y = \dfrac{e^{-x}}{1+x^2}$, show that

$$(1+x^2)^2 \frac{dy}{dx} + e^{-x}(x+1)^2 = 0.$$

(4 marks)

b) A curve is given by the parametric equations

$$x = \sin^2 t, \qquad \text{and} \qquad y = \cos 2t - t^2.$$

Find $\dfrac{d^2y}{dx^2}$ in terms of t.

(5 marks)

c) By taking natural logarithm both sides, differentiate

$$y = (1+x)^{\sin 3x}$$

with respect to x. Hence, evaluate $\dfrac{dy}{dx}$ when $x = 0$.

(6 marks)

QUESTION 3 (15 MARKS)

a) A ship is required to make a voyage of 200 km distance at a constant speed. When the speed of the ship is V km/hour, the cost is $\left(V^2 + \dfrac{4000}{V}\right)$ Ringgit Malaysia per hour. If

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}},$$

show that the total cost for the whole voyage is

$$C = 200V + \frac{800000}{V^2}.$$

Find the speed at which the ship should travel so that the total cost of the voyage is a minimum.

(6 marks)

b) The curve with equation $y = ax^3 + bx^2 + cx + d$ where a, b, c, and d are constants, passes through the points $(0, 3)$ and $(1, 0)$. At these points the curve has gradients -7 and 0, respectively.

i. Show that $a = -1$, $b = 5$, $c = -7$, and $d = 3$.

(2 marks)

ii. Show that the curve crosses the $x-$axis at the point $(3, 0)$.

(1 mark)

iii. Find the coordinates of the extremum points.

(4 marks)

iv. Sketch the graph of the curve and label all important points.

(2 marks)

QUESTION 4 (18 MARKS)

a) Evaluate each of the following integrals:

i. $\displaystyle\int (\cos 3x \cos 4x - \sin^3 2x)\ dx.$

(6 marks)

ii. $\displaystyle\int e^{-2x} \sin 2x\ dx.$

(5 marks)

b) Using the substitution $u = e^x + 1$, show that

$$\int \frac{e^x - 2}{e^x(e^x + 2)}\ dx = \int \frac{u - 3}{(u-1)^2(u+1)}\ du.$$

Hence, evaluate

$$\int \frac{e^x - 2}{e^x(e^x + 2)}\ dx.$$

(7 marks)

QUESTION 5 (10 MARKS)

a) Find the area of the region bounded by the curves $y = \dfrac{x}{2}$ and $y = \sqrt{x}$.

(4 marks)

b) Find the volume of the solid generated when the region bounded by $y^2 = 4x$ and $y = 2x - 4$ is revolved 360° about $y-$axis.

(6 marks)

QUESTION 6 (10 MARKS)

a) Solve the ordinary differential equation

$$e^{y^2}\frac{dy}{dx} = \frac{1}{x^2 y}.$$

(4 marks)

b) Solve the initial value problem given by

$$\frac{dy}{dx} + \frac{y}{x} = \frac{\ln x}{x^2}, \quad y(1) = 3.$$

(6 marks)

QUESTION 7 (17 MARKS)

a) A nonlinear equation

$$f(x) = x^4 - 2x - 3$$

has two roots, α, and β.

i. By setting $f(x) = 0$, show that the above nonlinear equation can be written in the form of

$$x = \left(\frac{3}{x} + 2\right)^{\frac{1}{3}}.$$

Hence, perform fixed point iteration method with the starting value $x_0 = 1.4$ to determine the first root, α. Use the stopping criteria $|x_i - x_{i-1}| < 0.0005$.

(5 marks)

ii. Use Newton-Raphson method with the starting value $x_0 = -0.5$ to determine the second root, β. Stop your iteration when $|x_i - x_{i-1}| < 0.0005$.

(5 marks)

b) Find an approximation for the value of

$$\int_2^6 \frac{x^2}{x-1}\, dx$$

by Simpson's rule using eight strips.

(7 marks)

LIST OF FORMULA

Trigonometric
$\cos^2 x + \sin^2 x = 1$
$1 + \tan^2 x = \sec^2 x$
$\cot^2 x + 1 = \mathrm{cosec}^2 x$
$\sin 2x = 2\sin x \cos x$
$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$
$\tan 2x = \dfrac{2\tan x}{1 - \tan^2 x}$
$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
$\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
$2\sin x \cos y = \sin(x+y) + \sin(x-y)$
$2\sin x \sin y = -\cos(x+y) + \cos(x-y)$
$2\cos x \cos y = \cos(x+y) + \cos(x-y)$

Logarithm
$a^x = e^{x \ln a}$
$\log_a x = \dfrac{\log_b x}{\log_b a}$

Differentiations
$\dfrac{d}{dx}[k] = 0,\ k$ constant
$\dfrac{d}{dx}[x^n] = nx^{n-1}$
$\dfrac{d}{dx}[e^x] = e^x$
$\dfrac{d}{dx}[\ln\lvert x\rvert] = \dfrac{1}{x}$
$\dfrac{d}{dx}[\cos x] = -\sin x$
$\dfrac{d}{dx}[\sin x] = \cos x$
$\dfrac{d}{dx}[\tan x] = \sec^2 x$
$\dfrac{d}{dx}[\cot x] = -\mathrm{cosec}^2 x$
$\dfrac{d}{dx}[\sec x] = \sec x \tan x$
$\dfrac{d}{dx}[\mathrm{cosec}\, x] = -\mathrm{cosec}\, x \cot x$

Integrations
$\int k\,dx = kx + C,\ k$ constant
$\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C,\ n \neq -1$
$\int e^x\,dx = e^x + C$
$\int \dfrac{dx}{x} = \ln\lvert x\rvert + C$
$\int \sin x\,dx = -\cos x + C$
$\int \cos x\,dx = \sin x + C$
$\int \sec^2 x\,dx = \tan x + C$
$\int \mathrm{cosec}^2 x\,dx = -\cot x + C$
$\int \sec x \tan x\,dx = \sec x + C$
$\int \mathrm{cosec}\, x \cot x\,dx = -\mathrm{cosec}\, x + C$

Fixed Point Formula
$x_{i+1} = g(x_i), \quad i = 0, 1, 2, \ldots$

Newton-Raphson Formula
$x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}, \quad i = 0, 1, 2, \ldots$

Trapezoidal rule
$\int_{x_0}^{x_N} f(x)\,dx = \dfrac{h}{2}\left[f_o + f_N + 2(f_1 + f_2 + \cdots + f_{N-1})\right],\ h = \dfrac{x_N - x_0}{N}$

Simpson's rule
$\int_{x_0}^{x_N} f(x)\,dx = \dfrac{h}{3}\left[f_o + f_N + 4(f_1 + f_3 + \cdots + f_{N-1}) + 2(f_2 + f_4 + \cdots + f_{N-2})\right],\ h = \dfrac{x_N - x_0}{N}$